Xgboost Classifier { Extreme Gradiant Boosting }

This is Dummy model → $\boxed{\text{Base model}}$ → probability = 0.5

{Dataset}

Salary	Credit	Approval	Residual
<=50 K	B	0	-0.5
<=50K	G	1	0.5
<=50K	G	1	0.5
>50K	B	0	-0.5
>50K	G	1	0.5
>50K	N	1	0.5
<=50 K	N	0	-0.5

Residual ← Calculated using this probability

λ → hyper parameter Setup using cross validation

1) Create a Binary Decision Tree using the feature
2) Calculate Similarity weight

$$\frac{\Sigma(\text{Residual})^2}{\Sigma(\text{Probability}(1-\text{Probability}) + \lambda}$$

Binary DT [-0.5, 0.5, 0.5, -0.5, 0.5, 0.5, -0.5]

Salary

<=50 | >50

[-0.50, 0.5, 0.5, -0.5] [-0.5, 0.5, 0.5]

→ Similarity weight = $\frac{[-0.5+0.5+0.5-0.5+0.5+0.5-0.5]^2}{0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)}$

$= \frac{0.25}{1.75} = \frac{1}{7} = 0.14$

⇓

Similarity weight $= \frac{[-0.5+0.5+0.5-0.5]^2}{0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+\lambda=0}$

$= 0$

Similarity weight $= \frac{[-0.5+0.5+0.5]^2}{0.5(1-0.5)+0.5(1-0.5)+0.5(1-0.5)+\lambda=0}$

$= \frac{0.25}{0.75} = \frac{1}{3}$

$= 0.33$

*** Let's say
$\lambda = 0$ for this Case
Usually λ value is 1

3> Calculate the Information gain

Info gain $= 0 + 0.33 - 0.14$
$= 0.19$

So based on highest info gain, feature is selected in a binary tree

4> Lets further go and split using next feature

Salary [-0.5, 0.5, 0.5, -0.5, 0.5, 0.5, -0.5]
Similarity weight = 0.14
info gain = 0.19

<= 50 [-0.5, 0.5, 0.5, -0.5]
Similarity weight = 0

> 50 [-0.5, 0.5, 0.5]
Similarity weight = 0.33

Credit → info gain $= 1 + 0.33 - 0$
$= 1.33$

B [-0.5]

G, N [0.5, 0.5, -0.5]

Similarity weight $= \frac{0.25}{0.5(1-0.5)} = \frac{0.25}{0.25} = 1$

Similarity weight $= \frac{0.25}{0.75} = \frac{1}{3} = 0.33$

→ like this will also happen in the next node. So based on info gain we choose the best

5> Now lets see how inferencing is happening.
in our dataset, let the first record will go to the base model

[<=50 B 0 -0.5] → $\boxed{\text{Base model}}$

$\log\left(\frac{p}{1-p}\right) = \frac{0.5}{0.5} = \log(1) = 0$
(for only base model)

Now this record is <=50, it will fall under <=50 branch. it is further divided as it is Bad (B).

$\sigma[0 + \alpha(1)]$ → learning rate $\boxed{\text{Usually } \alpha = 0.01}$
sigmoid

⇓

The o/p will be 0 to 1

Our Entire function look like this

$$\sigma\left[0 + \alpha_1(DT_1) + \alpha_2(DT_2) + \alpha_3(DT_3) + \dots + \alpha_n(DT_n)\right]$$

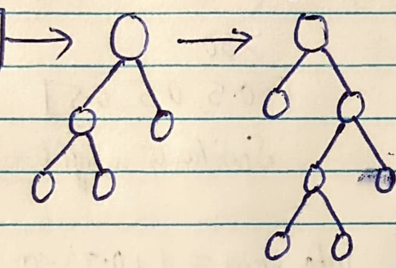
$$\Downarrow$$

This will be the final o/p when you trying to inference new record.

- It is very very difficult to calculate all these things Therefore it is also a black box model.

$$0 \quad + \alpha_1(DT_1) + \alpha_2(DT_2) + \dots + \alpha_n(DT_n)$$

$\boxed{\text{Base model}} \rightarrow$



Xgboost Regressor

$\boxed{\text{Base model}} \rightarrow$ 51 k

Exp	Gap	Salary (o/p)	Residual
2	Yes	40k	$-51+40=40-51=-11$ k
2.5	Yes	42k	-9k
3	No	52k	1k
4	No	60k	9k
4.5	Yes	62k	11k

$$\boxed{\text{Similarity weight} = \frac{\Sigma(\text{Residuals})^2}{\text{No. of Resi} + \lambda}}$$

[-11k, -9k, 1k, 9k, 11k]

$\boxed{\text{Exp}}$

≤ 2 → [-11] $\qquad$ >2 → [-9, 1, 9, 11]

$$S.W = \frac{(-11-9+1+9+11)^2}{5+1} = \frac{1}{6} = \cancel{28.8}\ 0.16$$

$$\lambda = 0 \rightarrow S.W = \frac{121}{1+0} = 121$$

$$\lambda = 1 \rightarrow S.W = \frac{121}{1+1} = 60.5$$

$$\lambda = 1 \rightarrow S.W = \frac{(-9+1+9+11)^2}{4+1} = \frac{144}{5} = 28.8$$

Information gain $= 60.5 + 28.8 - 0.16$
$= 89.13$

$\boxed{\text{Exp}}$ $S.W = 1/6$

≤ 2.5 $[-11, -9]$ | > 2.5 $[1, 9, 11]$

$S.W = \dfrac{(-11-9)^2}{2+1} = \dfrac{400}{3} \Rightarrow 133.33$

$S.W = \dfrac{(1+9+11)^2}{3+1} = \dfrac{441}{4} = 110.25$

Information gain $= 133.33 + 110.25 - 0.16$
$= 243.42$

- Now we gonna use this split as the info gain is higher than the previous split.

- Now lets do the inferencing

D.T 1

$\boxed{\text{EXP}}$ $S.W = 0.16$

≤ 2.5 $[-11, -9]$ $S.W = 133.33$ | > 2.5 $[1, 9, 11]$ $S.W = 110.25$

- now when ever any record do go to the base model for inferencing
  Let's say the record goes in ≤ 2.5 direction, now the average of $(-11, -9)$ is calculated $\dfrac{-11-9}{2} = 10$

$51 + \alpha_1(-10)$ similarly for >2.5 direction $51 + \alpha_1(7)$

- So for many Decision Trees

$O/P = 51 + \alpha_1(\overset{7}{-10}) + \alpha_2(D.T_2) + \alpha_3(D.T_3) + \dots + \alpha_n(D.T_n)$

- Again a black box model.